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Paleomagnetic Evidence for Rotation of the Italian Peninsula

WILLIAM LOWRIE

Institut für Geophysik, ETH-Hönggerberg, Zürich, Switzerland

WALTER ALVAREZ

Lamont-Doherty Geological Observatory of Columbia University, Palisades, New York 10964

The scaglia rossa, a pink pelagic limestone of mid-Cretaceous to middle Eocene age in the autochthonous, or parautochthonous, Umbrian Apennines offers an opportunity to evaluate paleomagnetically the rotational history of the Italian peninsula. Over 350 samples from 37 sites in this unit were collected during three field seasons in 1973–1974. Weak natural remanent magnetizations (6×10^{-8} to 8×10^{-6} G) had an initial soft component that could be removed by thermal demagnetization to 250°–350°C or by alternating field (af) demagnetization fields less than 100 Oe. The magnetic mineralogy is not simple. Where both magnetite and hematite are present, thermal demagnetization showed that the hematite magnetization is in the same direction as that of the magnetite, which indicates that hematization may have taken place shortly after deposition. AF treatment in both 100 and 200 Oe was used to clean each sample. Samples distributed geographically over the scaglia rossa depositional basin clearly showed the effect of a later rotation. The mean remanent vector for the formation is $D = 317°$, $I = 44°$, $\alpha_{95} = 6°$. Seventeen sites were sampled in the classic section at Gubbio, where paleontologic ages are well established. The inclinations vary by only a few degrees and are slightly low based on the present latitude of the sites, although they agree well with expected inclinations from a plate tectonics reconstruction. The declinations appear to show that the Italian peninsula was rotating during part of the 50 m.y. represented by the formation. The most probable interpretation shows a 45° counterclockwise rotation in the Campanian-Maestrichtian and a further 25° counterclockwise rotation after the middle Eocene. The pattern of magnetic reversals agrees well with the Cretaceous magnetic polarity time scale.

INTRODUCTION

Viewed in the light of plate tectonics concepts, the Mediterranean region is a zone of adjustment between the continental parts of the Eurasian and African plates. As a result of the relative movements of these two major plates since the Early Jurassic [*Pitman and Talwani,* 1972; *Smith,* 1971] a number of Mediterranean microplates have moved in a complex constantly changing pattern [*Dewey et al.,* 1973; *Alvarez et al.,* 1974]. The lines of evidence commonly used for accurate determination of plate motion (fit of continental margins, marine magnetic anomalies, transform faults) are not generally usable for the Mediterranean microplates [*Alvarez,* 1973]. Counterclockwise rotations of the Iberian peninsula, the Italian peninsula, Corsica, and Sardinia were first proposed by *Carey* [1958]. Subsequently, paleomagnetic evidence has been used to establish the counterclockwise rotation of Sardinia [*Zijderveld et al.,* 1970*a*], Corsica [*Nairn and Westphal,* 1968], Iberia [*Van der Voo,* 1969], the Southern Alps [*Zijderveld et al.,* 1970*b*; *Manzoni,* 1970; *Heller,* 1973], and the western Alps [*Westphal,* 1973].

Rotation of the Italian peninsula has been suggested [*Boccaletti and Guazzone,* 1970; *Smith,* 1971] on the basis of geological considerations. *Soffel* [1972], in a paleomagnetic study of the Euganean hills on the north side of the Po plain, found a 55° counterclockwise rotation between the Eocene and the Miocene. The Italian peninsula may have participated in this rotation, as Soffel suggested, but this cannot be assumed, since the Euganean hills and the Italian peninsula are separated by the Po basin, where several kilometers of Pliocene and Quaternary sediment accumulated and were strongly deformed [*Sestini,* 1970; *Mueller and Talwani,* 1971] in a depression that may mark a plate boundary.

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Geological background. Direct paleomagnetic study of the rotational history of the Italian peninsula has been hampered in the past by a lack of suitable formations. In the northern Apennines the partly volcanic eugeosynclinal sequences are found only in far-traveled nappes, while the autochthonous or parautochthonous, miogeosynclinal Umbrian and Tuscan sequences are entirely nonvolcanic. The central Apennines comprise a limestone sequence without volcanics and are entirely allochthonous, while the southern Apennines, in part volcanic, are also entirely allochthonous. The stable platform of Apulia and the Gargano peninsula in southeastern peninsular Italy exposes only carbonate rocks.

The Umbrian sequence begins with Triassic evaporites known only in wells. A massive shallow water limestone, the calcare massiccio, represents the lower Lias. Beginning in the middle Lias the carbonate platform broke up into a complex of seamounts and basins in which a sequence of largely pelagic limestones was deposited through the remainder of the Mesozoic. A gradually increasing terrigenous component in the Paleogene produced a transition from limestone through progressively more argillaceous marls until the onset, in the middle Miocene, of the turbidite deposition that produced the marnoso arenacea flysch. A 50-m.y. interval is represented by the Turonian to middle Eocene scaglia rossa, a pink pelagic limestone in the autochthonous or parautochthonous Umbrian sequence. The stratigraphy of the Umbrian sequence has been discussed by *Bortolotti et al.* [1970], and the tectonics of the region by *Centamore et al.* [1972].

The aims of this study of the paleomagnetism of the scaglia rossa were (1) to determine whether or not there has been a rotation of the Italian peninsula relative to stable Europe and if so to find approximately when the rotation occurred and (2) to resolve whether the arcuate trend of the northern Appennines (Figure 1) was due to oroclinal bending or is the result of

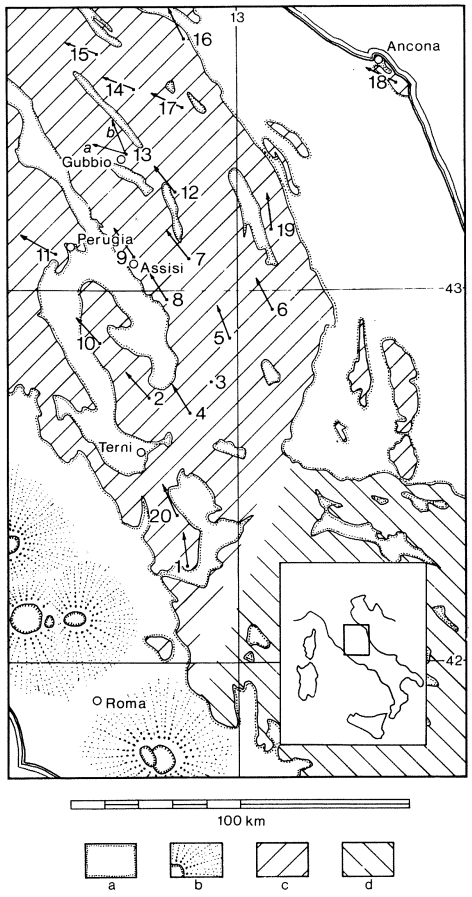


Fig. 1. Map of the outcrop area of the scaglia rossa in the northern Apennines, showing the sampling site locations. The arrows represent the site mean directions of the horizontal component of magnetization. Sixteen sites were placed at locality 13 near Gubbio; 13*a*, mean of six sites below the interval upper Campanian–lower Maestrichtian; 13*b*, mean of six sites above this interval (see Table 3). Pattern *a* is the upper Tertiary–Quaternary sedimentary cover; *b*, Quaternary volcanic centers; *c*, Umbrian sequence (Mesozoic–lower Tertiary); *d*, Abruzzi sequence (Mesozoic–lower Tertiary). Map is modified from *Lowrie and Alvarez* [1974].

a progressive change in orientation of anticlinal axes along the trend of the range.

Sampling. Field collections of paleomagnetic samples were made in the summer and autumn of 1973 and supplemented by an additional collection in early 1974. The paleomagnetic samples were drilled in the field by using a portable gasoline-powered diamond drill and oriented in situ by using a magnetic compass. At a few sites in the Bottaccione gorge at Gubbio it was necessary to drill through wire screening, which is used extensively in Italy to prevent rockfalls. Samples from these sites were oriented by sighting from across the road to an orienting device mounted in the drill hole. This method was found to be just as accurate as orienting a conventional device in the drill hole.

The scaglia rossa has been used in medieval and modern construction as a popular building stone because of its attractive appearance and the ease with which it can be worked. There are many quarries in the formation and many road cuts where it is exposed. Extensive sampling was possible at 20 localities spread over the geographical range of the formation (Figure 1) and additionally at 17 sites spaced stratigraphically in a single well-exposed section through the entire scaglia rossa.

From 7 to 18 samples were collected at each of the 37 sites, for a total of 351 samples. Each gave from (mostly) one to (occasionally) three specimens in the form of right circular cylinders, 2.5 cm diameter and 2.5 cm high. These were investigated to establish the paleomagnetic and rock magnetic properties of the scaglia rossa.

REMANENT MAGNETIZATIONS

Intensities. A major difficulty with using limestones for paleomagnetic studies has been the extreme weakness of their

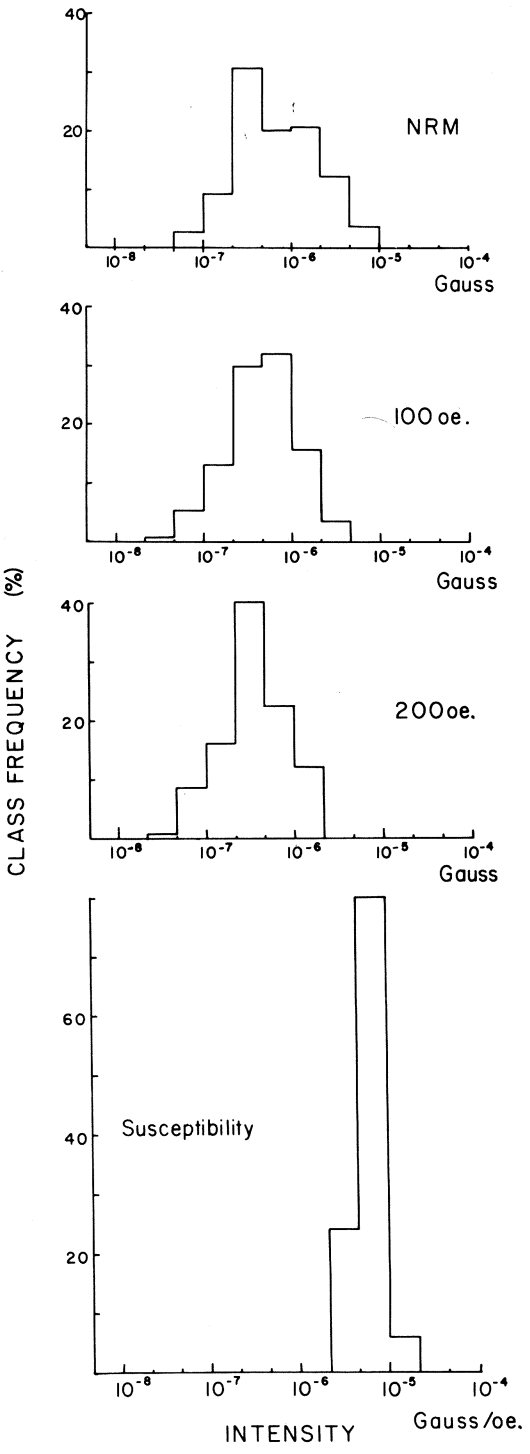


Fig. 2. Histograms of the NRM intensities, the remanent intensities after af demagnetization in 100 Oe and 200 Oe, and the bulk susceptibilities of 351 samples of the scaglia rossa.

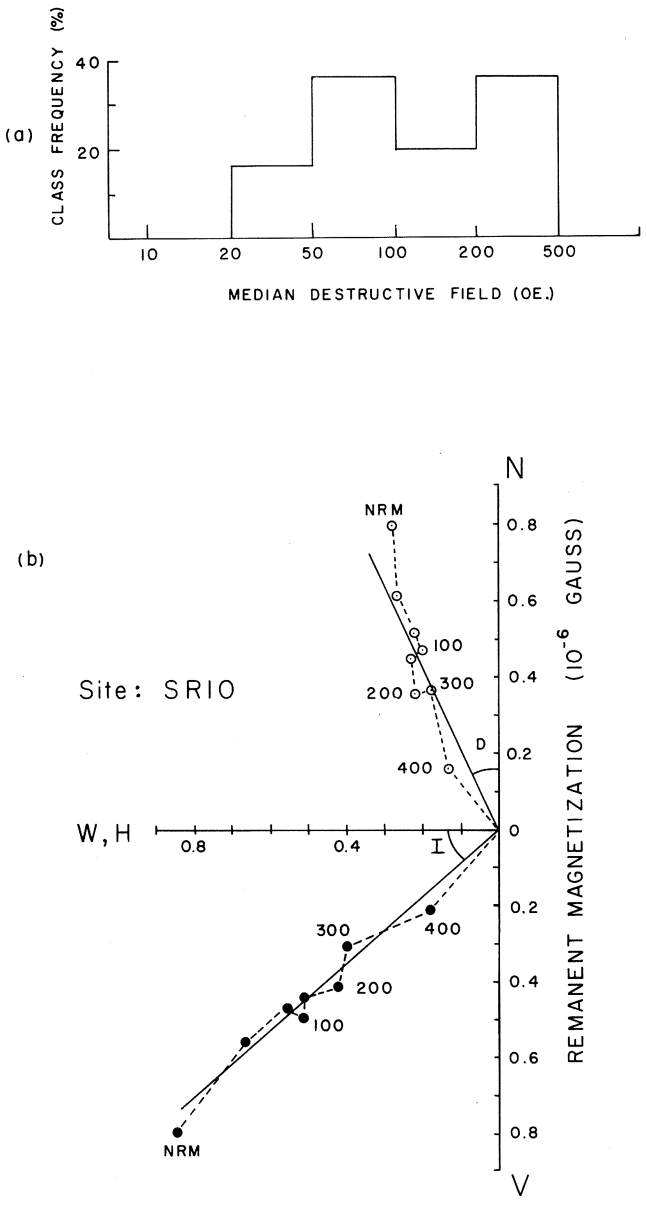


Fig. 3. (*a*) Histogram of MDF of representative specimens from 37 sites. (*b*) Representative vector diagram illustrating the stability against af demagnetization of a sample from site SR10.

remanent magnetization, which often was below the sensitivities of most magnetometers. This obstacle has been removed by the development of sophisticated magnetometers such as the Digico complete results magnetometer [*Molyneux,* 1972], which was successfully used in an earlier limestone study [*McElhinny and Opdyke,* 1973] and greatly facilitated analysis of the scaglia rossa.

The intensities of natural remanent magnetization (NRM) were very weak, ranging from 5.3×10^{-8} G at site 13A to 7.7×10^{-6} G at site 8. After demagnetization in a peak alternating magnetic field of 200 Oe the remanent intensities were further reduced; they then ranged from 4.3×10^{-8} G to 1.9×10^{-6} G. The widely dispersed remanent intensities followed an approximately log normal distribution at each stage of demagnetization (Figure 2). The susceptibilities were also very weak but were more uniform than the remanences, ranging from 2.1×10^{-6} G/Oe to 1.2×10^{-5} G/Oe. In spite of the weakness of the remanences they were quickly measured and gave reliable data. A remanence of intensity 6.3×10^{-7} G could be measured with less than 2% standard error in intensity and with directional repeatability within 1°.

Stabilities. One specimen from each site was subjected to alternating field (af) demagnetization in progressively increasing peak fields. Each NRM contained a significant soft component that was easily removed by low alternating fields. The median destructive fields (MDF) ranged from 34 Oe to 480 Oe; almost half were less than 100 Oe (Figure 3*a*). The distribution is bimodal. Although there is not a one-to-one correspondence, the MDF is loosely associated with the pink coloration; the pinkest samples generally have the highest MDF, and the whitest have the lowest MDF. In all cases, however, the MDF showed that at least half of the NRM is carried by coercivities of only a few hundred oersteds or less. An average of 40% of the NRM was erased by 100-Oe af, and 70% of the NRM was erased by 200-Oe af.

After removal of the initial unstable component, each remanence was quite stable against further af demagnetization. In peak fields of more than 600 Oe, anhysteretic remanent magnetization (ARM) became an increasingly serious factor which made it unpractical to continue demagnetizing in higher fields. Clearly, the stability of the NRM is such that it can be cleaned to give a stable remanence that persists for a few hundred oersteds against af cleaning. The inclination and declination of the stable remanent vector were clearly defined for each test specimen (Figure 3*b*).

Stability tests [*Briden,* 1972; *Symons and Stupavsky,* 1974] were applied to the demagnetization behavior of the representative sample for each site to determine the optimum field for blanket demagnetization of the remaining specimens from the site. Generally this field lay in the range 100–200 Oe.

Site statistical analysis. All samples were demagnetized in 100 Oe and 200 Oe. The directions at each site were tested, after corrections were made for bedding, to see if the azimuths were uniformly distributed and to test whether the angular deviations from the site mean conformed to a Fisher distribution [*Fisher,* 1953]. At some of the sites, few data were available, invalidating these tests, which cannot reasonably be applied to a small number of samples. Although the radius of the circle of confidence (α_{95}) may not have its true meaning under these circumstances, it is a useful guide to the dispersion of the data.

The vector mean directions of magnetization were computed for every site at the NRM, 100-Oe and 200-Oe demagnetization stages. The overall scatter in the data was reduced by correcting for the bedding attitudes at each site. *Fisher*'s [1953] statistics were used to derive the radius of the circle of confidence of the mean, α_{95}. Site mean values for NRM intensity, susceptibility, and the site mean directions and site statistics are shown in Table 1. The circles of confidence were generally small after the initial demagnetization; at 100 Oe the average value of α_{95} was 9.3°, and at 200 Oe it was 9.2°.

The optimum direction for each site was taken to be the site mean with minimum angular dispersion. The representative site mean directions were nonuniformly distributed about their mean (Figure 4), but the distribution of angular deviations from the mean was Fisherian. In determining the formation mean, data from three sites (SR3, SR3A, and SR13K) were rejected for reasons discussed below. Reliable paleomagnetic data from the other 34 sites were used to determine the formation mean direction for the scaglia rossa, which was as follows: declination, 317.2°; inclination, 43.7°; $\alpha_{95} = 6.1°$.

TABLE 1. Summary of Site Mean Remanent Magnetic Properties

Site	N	NRM, 10^{-6} G	k, 10^{-6} G/Oe	NRM			100 Oe			200 Oe		
				D	I	α_{95}	D	I	α_{95}	D	I	α_{95}
1	8	0.815	6.53	6.2	71.2	7.6	*355.0*	*41.4*	*7.6*	353.7	44.7	7.9
2	7	0.451	7.00	86.7	15.2	42.0	*128.0*	*-42.1*	*6.2*	137.6	-43.3	8.5
3	8	1.193	7.88	18.3	30.5	6.1	0.2	33.0	8.6	4.0	27.1	9.2
3A	7	0.789	3.85	37.3	60.1	8.9	44.8	46.7	14.7	41.3	53.4	6.8
4	7	1.268	6.97	331.3	47.8	3.4	*329.7*	*40.5*	*5.9*	329.4	41.5	9.2
5	8	0.374	5.26	331.0	23.2	12.8	*336.6*	*24.7*	*13.3*	339.7	26.1	13.5
6	8	3.805	6.80	335.3	59.6	11.5	*352.8*	*42.6*	*9.7*	343.6	41.4	16.1
7	10	3.493	9.03	35.7	79.5	11.5	326.1	45.9	12.1	*324.6*	*45.0*	*9.6*
8	8	3.763	7.66	306.4	44.1	4.4	329.0	45.9	8.1	*328.2*	*40.4*	*7.3*
9	12	0.319	4.49	159.0	-39.1	4.2	*149.1*	*-47.9*	*3.1*	146.7	-47.0	4.2
9P	7	0.406	4.74	160.2	-39.5	4.9	*153.7*	*-46.8*	*3.9*	151.9	-47.6	4.2
9W	5	0.196	4.13	157.3	-38.6	9.5	*142.4*	*-49.1*	*4.6*	139.7	-45.7	8.4
10	7	0.940	4.85	329.4	43.7	6.3	*321.0*	*40.9*	*5.9*	318.4	41.0	6.8
11	12	0.684	4.93	287.8	41.0	11.5	288.8	19.6	11.0	*298.8*	*37.3*	*9.1*
12	7	5.977	9.39	286.8	53.0	6.7	318.2	49.1	11.6	*322.7*	*44.8*	*10.9*
13A	13	0.152	5.97	337.7	45.0	17.2	294.6	48.2	24.1	*329.3*	*41.5*	*10.0*
13B	10	0.487	7.74	328.3	34.9	10.8	305.2	35.3	10.8	*333.0*	*38.5*	*6.6*
13C	11	1.215	6.92	331.4	41.1	11.2	293.4	33.2	10.6	*307.4*	*39.1*	*9.3*
13D	11	0.279	4.90	346.4	48.2	12.4	281.7	37.5	7.4	*282.5*	*33.4*	*6.7*
13E	10	0.380	5.26	353.5	48.5	10.3	*290.3*	*36.3*	*9.5*	289.2	42.4	10.7
13F	10	0.177	5.17	140.4	-17.8	21.6	*168.6*	*-33.7*	*16.5*	171.0	-30.7	17.2
13G	10	0.426	5.62	132.6	-30.1	10.3	*154.8*	*-37.8*	*6.3*	142.3	-37.9	8.2
13H	9	0.927	6.65	342.1	31.4	7.9	319.1	31.1	7.3	*328.0*	*33.5*	*7.3*
13I	10	0.197	3.91	113.6	-24.7	8.9	*141.0*	*-34.3*	*7.9*	143.2	-37.4	9.2
13J	10	0.346	4.76	318.6	45.5	4.9	284.8	39.0	3.9	*295.7*	*38.7*	*3.9*
13L	10	0.214	5.86	108.4	-20.5	29.8	171.1	-34.8	16.7	*174.4*	*-36.0*	*14.8*
13M	8	0.134	5.89	87.0	-9.5	10.0	*153.6*	*-41.5*	*10.6*	143.0	-40.6	12.4
13N	8	0.250	6.40	55.6	-21.3	22.3	59.3	-63.1	51.7	*110.9*	*-52.1*	*14.1*
13O	8	0.461	5.35	335.8	61.1	12.1	*276.6*	*43.2*	*11.6*	284.8	37.9	12.1
13P	8	0.376	5.85	320.8	54.8	6.8	274.0	32.5	7.7	*279.5*	*35.5*	*7.1*
13S	14	0.265	5.71	285.1	56.9	13.7	250.6	39.1	12.2	*261.3*	*43.5*	*12.1*
14	8	1.610	7.60	259.1	48.3	5.1	*287.8*	*47.5*	*4.5*	290.7	51.0	5.6
15	13	4.466	9.69	258.1	66.8	12.0	307.2	64.2	9.8	*295.3*	*61.0*	*9.6*
16	8	1.081	6.77	329.0	62.2	5.2	*317.7*	*47.1*	*4.9*	333.2	50.4	9.8
17	8	1.633	7.77	292.8	50.3	4.7	293.4	43.8	5.2	*294.8*	*42.8*	*4.8*
18	8	1.114	7.34	295.6	49.8	4.5	*288.8*	*54.4*	*9.0*	283.7	52.2	11.0
19	7	0.695	5.91	343.9	54.7	6.1	*345.3*	*47.1*	*5.0*	353.4	48.0	5.6
20	7	0.276	7.51	350.1	42.7	21.6	*348.3*	*33.2*	*16.9*	335.5	41.9	18.0

N is the number of samples measured at each site. The NRM intensity and the bulk susceptibility k are simple arithmetic mean values, since there was little within-site variation in these parameters at most sites. The vector mean directions (D, declination; I, inclination) and the radius of the circle of 95% confidence (α_{95}) are shown for the NRM and the magnetizations after af demagnetizations in 100 Oe and 200 Oe. The values in italics are the directions used in arriving at the overall mean direction of magnetization of the scaglia rossa.

MAGNETIC MINERALOGY

The coloration of the scaglia rossa varies from dark red to delicate pink, and in some places the color is off-white. Although we have done no chemical analysis to determine the origin of the coloration, the color of red sediments is often attributed to pigmentary hematite. If an NRM is carried by hematite, it is doubtful that it can be a primary remanence, since hematite probably forms either (1) by oxidation of the primary magnetic minerals or (2) by precipitation in situ from percolating groundwater. In the first of these possibilities, part of the original remanence may still remain, and in the second the secondary remanence may be superposed on a depositional remanence carried by detrital magnetic minerals. Consequently, the directions of magnetization measured in the scaglia rossa may be a combination of primary and secondary magnetization vectors. It is therefore important to establish the significance of secondary magnetization in the scaglia rossa.

Several types of magnetic mineralogical evidence were obtained: Curie temperature analysis of separated magnetic minerals, isothermal remanent magnetization (IRM) acquisition curves, thermal demagnetization curves, and comparison of the directions of magnetization in white and pink samples from the same bed.

Analysis of extracted magnetic minerals. Magnetic minerals were extracted from the limestone by a method previously used for deep-sea sediments [*Løvlie et al.*, 1971]. A buffered solution of acetic acid was used to break down the limestone matrix. The process was slow but effective. The residue was passed several times in a slow vertical flow of water through a tube between the poles of an electromagnetic separator. The extract was dried for thermomagnetic analysis. Because of the large amount of rock required to extract a useful amount of magnetic minerals it was only possible to derive extracted minerals at sites 3A, 6, and 11, where large blocks had been collected in addition to the drilled samples.

The estimated concentrations by weight of the extracts were from 0.02% to 0.16% (Table 2), similar to the magnetic mineral concentrations found in modern deep-sea sediments [*Løvlie et al.*, 1971]. As in the latter situation, the extraction technique was only partially effective in removing the ferromagnetic grains, yet it also removed a large number of light-colored detrital grains that may be associated with included ferromagnetic grains. The concentration estimates are there-

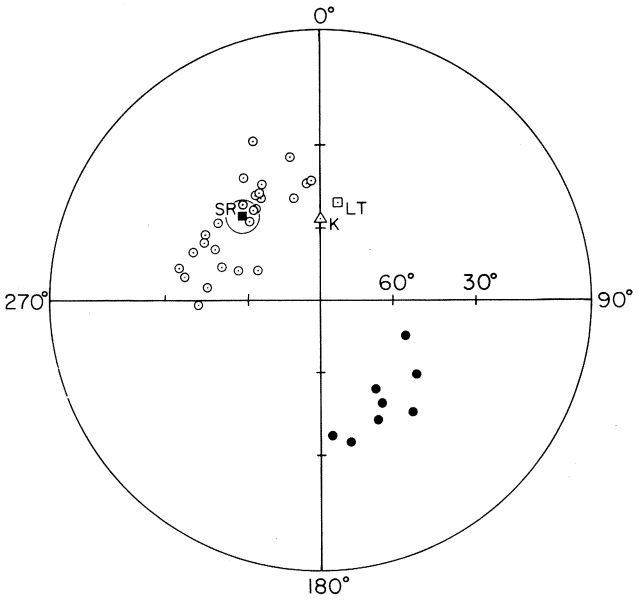


Fig. 4. Lower-hemisphere stereogram showing the acceptable site mean directions of 34 sites in the scaglia rossa. Open (closed) circles represent normally (reversely) magnetized sites; the encircled solid square represents the formation mean direction and the circle of 95% confidence, while the open square and triangle represent the directions of the mean lower Tertiary and Upper Cretaceous fields, respectively, at the present site coordinates.

fore not believed to be accurate, although they do reflect the very low concentration of the detrital fraction at each site.

Thermomagnetic analysis was carried out to 700°C in air in a field of 4000 Oe on three specimens of the extract from each site (Table 2). At sites SR3A and SR6 the thermomagnetic curves showed single well-defined Curie points near 540°C and 550°C, respectively (Figure 5). The magnetic mineral at these sites corresponds to magnetite, perhaps with minor amounts of titanium in the lattice. Although no hematite Curie point was discernible in the curves from these sites, this does not exclude its presence in smaller amounts than could be detected. At site 11 the largest amount of extract was obtained, but it was only weakly magnetic. The thermomagnetic curves were noisy, and only one curve gave usable data. This curve showed a Curie point at 640°C and a less well defined Curie point at 580°C, corresponding to hematite and magnetite, respectively. At this site the hematite is probably in greater proportion to the magnetite than at the other two sites.

The thermomagnetic analyses of the extracted magnetic minerals indicate that magnetite and hematite are present in the scaglia rossa, probably in varying proportions at different sites.

IRM acquisition curves. *Dunlop* [1972] has shown that the acquisition curves of IRM may be used to obtain an improved understanding of the magnetic mineralogy of rock samples. The ability to af demagnetize rocks is usually limited by apparatus design to peak fields of no more than 2,000–3,000 Oe, and thermal demagnetization may cause irreversible changes in the magnetic minerals. Acquisition of IRM permits an examination of the total coercivity spectrum of the sample, in particular the high coercivity fraction, and does not alter the magnetic mineralogy.

Successive IRM's were given in progressively increasing fields to a representative specimen from each site by using a 4-inch electromagnet to provide fields up to 10 kOe. Remanence components harder than 5 kOe may safely be attributed to hematite, probably present as the pigment which gives the scaglia rossa its distinctive color. Coercivities between 0 and 5 kOe may be due to specular hematite [*Dunlop,* 1972]. The coercivity of an elongate, single-domain, rod-shaped grain of magnetite or maghemite is around 1500 Oe, and magnetization components softer than this may be attributed to either of these two minerals. As maghemite is metastable, inverting to hematite under oxidizing conditions, it is likely that magnetization components softer than 1500 Oe in the scaglia rossa are due to magnetite grains.

We follow the example of *Dunlop* [1972] by assigning rather arbitrarily the approximate relative IRM contributions of magnetite, specularite, and pigment to three coercivity ranges: <1 kOe, 1–3 kOe, and >3 kOe, respectively. The coercivity spectra were simply obtained by numerical differentiation of the IRM acquisition curves (for details, see *Dunlop* [1972]).

The IRM acquisition curves fall into three distinct groups (Figure 6*a*). The first consists of curves which level off at applied fields of around 1000 Oe, the second group shows little or no tendency to level off in fields up to 10 kOe, and the third shows a combination of both these effects. As emphasized by the coercivity spectra for representative samples from each group (Figure 6*b*), the first group is dominated by coercivities lower than 1 kOe. The dominant magnetic mineral in these samples, representing 9 sites, is probably magnetite. The coercivities of the second group are distributed more widely

TABLE 2. Concentrations and Curie Points of Magnetic Extracts From Sites 3A, 6, and 11

Site	Magnetic Extract Concentration, %	Curie Point, °C
SR3A	0.02	540 ± 10
SR6	0.07	550 ± 10
SR11	0.16	580, 640

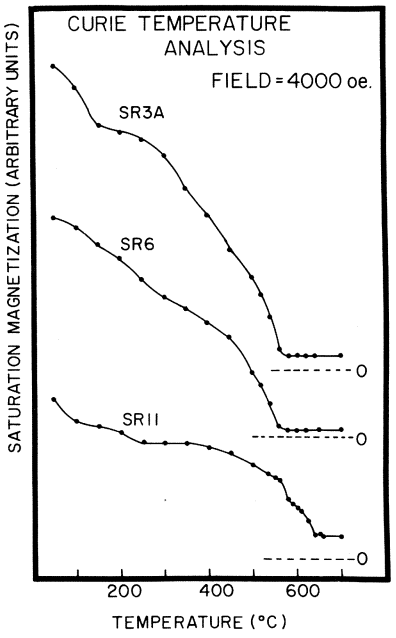


Fig. 5. Curie temperature analysis of representative specimens from three sites.

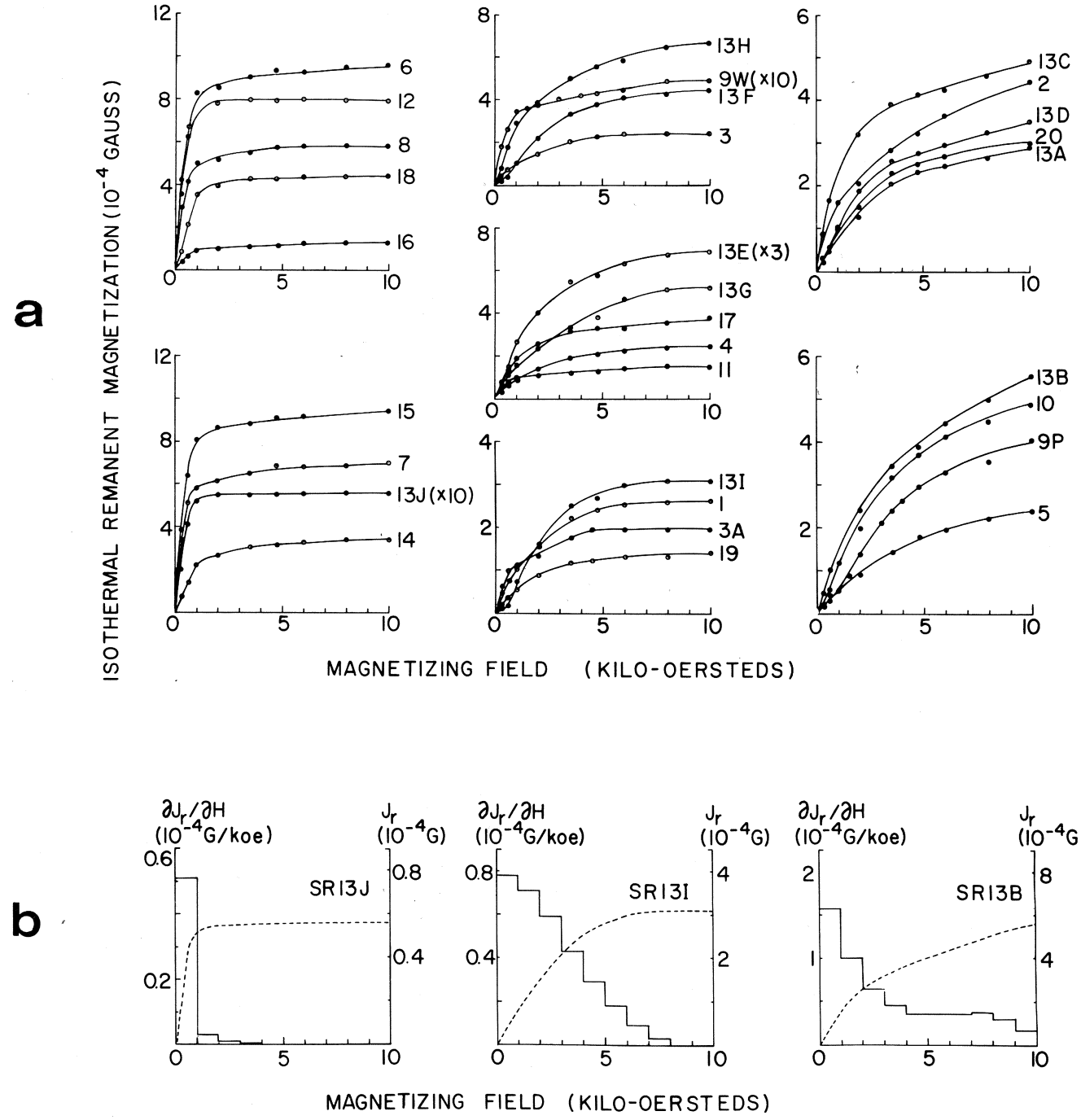


Fig. 6. Isothermal remanence acquisition curves for representative samples from each site; a pink (9P) and a white (9W) sample from SR9 are both shown.

between 0 and 5 kOe; the spectrum approaches zero as saturation remanence is reached in fields above 5 kOe. It is very likely that the harder magnetic mineral in the 13 sites which showed this behavior is specular hematite. At the remaining 9 sites the IRM acquisition curves do not reach saturation in 10 kOe. The hardest part of the remanence may be attributed to pigmentary hematite. However, the character of many IRM curves indicates that the magnetic mineralogy consists of both magnetite and hematite.

Low-temperature analysis. *Nagata et al.* [1964] proposed a simple test for investigating the magnetic mineralogy of rock samples based upon their low-temperature characteristics. The test depends upon the change in remanent magnetization associated with the Morin transition in hematite at −23°C and a similar change accompanying the vanishing of the principal magnetic anisotropy constant K_1 of magnetite at −155°C. The temperatures at which both transitions occur are depressed by the presence of titanium in the crystal lattice, and both are suppressed entirely if the magnetic grains are single-domain size.

Following the method of *Nagata et al.* [1964], strong field (5-kOe) IRM was given at room temperature to selected pink and white scaglia rossa specimens. Each was then immersed in liquid nitrogen until it reached thermal equilibrium. At liquid nitrogen temperature (−193°C) the specimen was again given saturation IRM. The variation of the magnetization as the pecimen warmed to room temperature was monitored con-

tinuously. No changes indicative of either transition were observed.

The low-temperature analyses did not help further in identifying the magnetic minerals. The absence of any magnetic transitions shows that if magnetite or hematite were present, as suggested by the other magnetic mineralogy evidence, their transitions were being suppressed and consequently their grain sizes must lie in the single-domain range.

Thermal demagnetization. For NRM components carried by hematite grains, thermal demagnetization is more effective than af demagnetization because of the very high coercivities associated with single-domain hematite grains. Specimens from every site were heated to successively higher temperatures to a maximum of 700°C. Cooling between stages was in a region where the geomagnetic field was cancelled with Helmholtz coils to less than 50 gammas. Thermoremanent magnetization (TRM) components resulting from improper field cancellation were always very obvious when present, so that allowance could easily be made in interpreting the thermomagnetic data.

Most samples showed a pronounced change in intensity on first heating (Figure 7) which consisted of an initial increase for reversely magnetized samples and a decrease for normally magnetized samples, reflecting the removal of a soft normal component of magnetization. The intensity and direction of the remanence stabilized in most cases after heating to 250°–350°C. The remanence decayed uniformly from there to 550°C, with slight directional scatter in most cases. Between 550°C and 600°C there was a small, but more abrupt, intensity decrease by about a half in many cases. After heating to above 600°C the directional scatter increased and gave random magnetization directions in many of the samples. However, in some samples the directions were consistent to above 650°C. These curves revealed the thermal unblocking of NRM components carried by grains of magnetite and hematite, with both minerals present at many sites. However, the superposed hematite contribution always was small in relation to that of the magnetite.

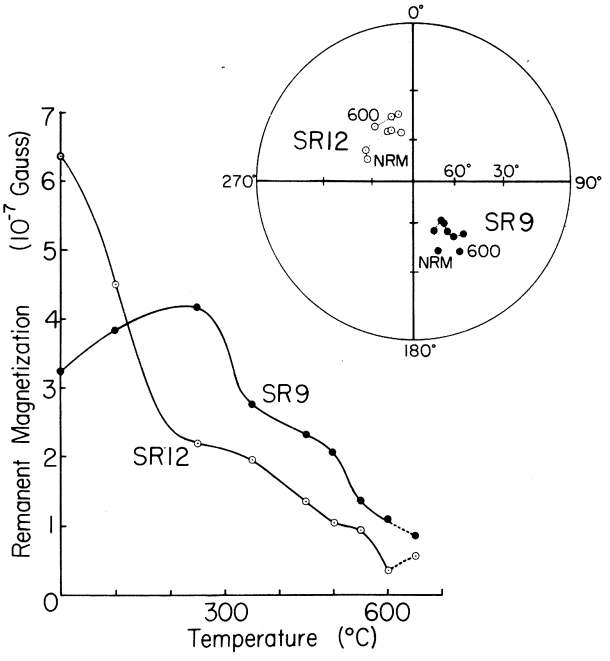


Fig. 7. The effect of thermal demagnetization on the intensity and direction (inset is a stereographic projection) of representative normal samples (SR12; solid points) and reversed samples (SR9; open circles).

The most interesting results with respect to interpreting the remanent magnetic properties of the scaglia rossa were obtained from those sites where IRM acquisition curves also indicated the presence of both magnetite and hematite (Figure 6). Two of 12 thermal demagnetization curves for these sites were discarded because of spurious TRM components acquired during cooling. The mean directions of the stable magnetization component below 550°C and the magnetization remaining after heating to 600°C and 650°C were measured, and the angular discrepancy between them was calculated. The mean value of this angular discrepancy at the 10 sites was 18° (standard error, 5°). If the direction of the remanent magnetization is assumed to be due to (1) a combination of approximately equal proportions of magnetite and hematite below 550°C and (2) hematite alone above this temperature, then the thermal demagnetization results do not indicate any large directional difference between the two components.

The implication of the thermal demagnetization results is that the hematization in most samples resulted in a secondary magnetization component in essentially the same direction as the stable primary remanence.

Comparison of magnetizations of pink and white samples. At site 9, near Assisi, five white and seven dark pink samples were collected from differently colored sections of a single bed. The IRM acquisition curves of one of the pink samples showed that only a small part of the coercivity spectrum lay below 1000 Oe, while 80% of saturation IRM was achieved in the same field in a white sample. All of the samples were af demagnetized, and optimum directions were found to result from 100-Oe treatment.

The mean direction of the pink samples was $D = 153.7°$, $I = -46.8°$, and $\alpha_{95} = 3.9°$; that of the white samples was $D = 142.4°$, $I = -49.1°$, and $\alpha_{95} = 4.6°$. The angle between these mean directions is 7.9°. Although each mean lies outside the circle of 95% confidence of the other, the circles of confidence overlap. It is not safe to attribute too much significance to α_{95} values for such small numbers of data, but the data show that there is no large directional difference between the two sets of magnetizations.

This is an especially important result because of the reversed direction of magnetization at site 9, which (like other reversed sites in the scaglia rossa) is almost antipodal in direction to the normally magnetized sites (Figure 4).

Two interpretations are possible, depending on how the NRM is carried. Acquisition curves of IRM indicate the total spectrum of coercivities in a sample but not the fraction carrying the NRM. For example, although the pink samples were apparently dominated by hematite, this did not necessarily contribute significantly to the NRM which could be carried by a small original magnetite fraction.

The color change within a single bed was abrupt but did not allow us to identify whether the white section represented an originally pink bed that was later bleached. If this was not the case, the presence of both pink and white colors in the same bed implies a postdepositional origin for the pink color possibly related to precipitation from percolating groundwater or from water motion during compaction and dewatering of the sediment. This situation may have occurred a considerable time after deposition.

The closeness of the directions of pink and white samples and their reversed polarities argue in this case that the later hematization did not contribute significantly to the NRM of the pink samples. A similar argument may be made in the case that the white section represents a leaching effect.

If the hematite does contribute significantly to the NRM, the fact that both pink and white samples have similar reversed directions strongly suggests that the secondary magnetization was acquired soon after the primary magnetization. It is possible that it could have been acquired considerably later, during a subsequent period of reversed geomagnetic polarity, but this is a less likely explanation.

Either interpretation of the data from site 9 leads to the conclusions (1) that the primary magnetizations have not been appreciably altered in direction and (2) that the remanence is sufficiently contemporaneous with the deposition for magnetic stratigraphy to be valid.

Summary of magnetic mineralogy evidence. The results of the magnetic mineralogy studies in the scaglia rossa are consistent with the view that some of the pelagic limestone samples contain very fine grained magnetite, some contain very fine grained hematite, and most contain both. Considering the ease with which the bulk of the NRM was erased by only a few hundreds of oersteds at every site, most of the NRM did not appear to be carried by single-domain hematite whose coercivities are several thousands of oersteds. The results indicate that hematization occurred at many sites, possibly soon after the acquisition of the primary remanence. It did not completely replace the original magnetite and did not appreciably alter the NRM direction.

The origin of the NRM of pelagic limestones is not yet completely understood. Early studies of acquisition of depositional remanent magnetization (DRM) by redepositing varved sediments [*King,* 1955] showed that an inclination error existed which made the varves unreliable for paleomagnetic studies if the error was not corrected. It has since been shown that modern deep-sea sediments do not have an inclination error but record faithfully the direction of the geomagnetic field [*Opdyke and Henry,* 1969] or the laboratory field [*Kent,* 1973]. The NRM is evidently acquired in deep-sea sediments by a different mechanism that that involved in DRM.

A postdepositional mechanism for remanence acquisition in wet sediments was advocated by *Irving and Major* [1964]. This postulated the alignment of grains of magnetic mineral within pore spaces of the sediment parallel to the ambient field. *Kent* [1973] has shown that this is a viable mechanism of acquisition of NRM in some deep-sea sediments, which preserves accurately the direction of the field. If this phenomenon is widespread in deep-sea sediments, the remanent magnetization is acquired shortly after the time of deposition.

The primary remanence of the scaglia rossa was probably acquired in this way. The effects of compaction and cementation on the magnetization of older, and perhaps deeply buried, deep-sea sediments are not known. In the absence of data pertaining to these effects we conclude that the formation probably carries a primary remanence whose directions may reasonably be assumed to approximate closely the ambient geomagnetic field directions at, or soon after, the time of deposition. The paleomagnetism of the formation is regarded as a valid indicator for paleomagnetic-tectonic study and to be sufficiently contemporaneous with deposition to give the scaglia rossa a useful magnetic stratigraphy.

Results of Geographically Distributed Sampling

Two tectonic questions were addressed by this part of the study. These were (1) whether the Italian peninsula has rotated relative to Europe and (2) whether the curvature of the Umbrian Apennines was produced by oroclinal bending of the mountain chain about a vertical axis or is due to a north to south change in the orientation of the fold axes. The broad geographic distribution of the 20 localities sampled over an area 100 km × 150 km was designed to reduce the importance of local structural complications and to resolve the question of possible bending of the Apennine chain. The trend of the folds changes by 80° (from N55°W in the north to N25°E in the south) within the area sampled. If an originally straight chain had been subsequently bent, the resulting 80° difference in paleomagnetic declination would be easily recognizable.

General stratigraphy and paleontological dating. The depositional conditions represented by the scaglia rossa were very long-lived, persisting about 50 m.y. from the Turonian to the middle Eocene. Three members are informally recognized: pink limestone with red chert forms the upper and lower members, and a chert-free pink limestone forms the middle member. These members have been dated paleontologically at a few sections, but because of the lack of detailed stratigraphic studies it is not known whether the boundaries are significantly time transgressive.

It is often difficult to determine the age at a particular site. Complete, well-exposed, unfaulted sections through the entire scaglia rossa are rare, and selection of a site is generally dictated by the vagaries of outcrop and the presence of unfractured rock suitable for drilling. Absence of chert identifies the middle member, and it is usually possible to distinguish the upper from the lower cherty member, although local faults and folds, sometimes masked in covered intervals, may make this impossible.

Paleontological dating has been undertaken with disappointing results, despite the fact that the scaglia rossa is a pelagic limestone rich in microfossils. The limestone is generally hard and well cemented, these characteristics making it impossible to extract individual tests. Softer, marly beds are often decalcified, the fossils being destroyed.

Samples from 16 sites were examined for nannofossils. The coccoliths were difficult to obtain and almost invariably overgrown and poorly preserved. A further difficulty is that in many sites there has been reworking of earlier fossils. In view of the small amount of material obtainable the site age must be taken to be at least as young as the youngest fossil identified.

Foraminifera were extracted from only a few samples; frequently the limestone is rich in foraminifera, which must usually be studied in thin section. This work, still in progress, will give improved age determinations for the sites, although critical surface details of the tests may be unrecognizable in thin section. Indicative of the difficulty is that *Renz* [1936], studying the foraminifera of the scaglia rossa in thin section, was unable to identify the boundaries between the Turonian, Coniacian, Santonian, and Campanian. No radiolaria were found in any of our samples.

Detailed discussion of the paleontological information available and of the relationship between remanent magnetization and age will be deferred until the results of study of the foraminifera in thin section are available.

Interpretation of paleomagnetic data. After scrutinizing closely the paleomagnetic directions of the widely distributed sites in the scaglia rossa it was necessary to reject the data from sites SR3 and SR3A. The directions of magnetization of these two sites disagreed strongly with the rest of the scaglia rossa data and with each other. They were located on opposite flanks of what may be an ancient seamount. (*Centamore et al.* [1971] have identified ancient seamount and basin topography elsewhere in the northern Apennines.) The geology of the sites

was complicated by indications of severe local deformation, with badly shattered rock and many small faults. These local tectonic problems made interpretation of the discordant paleomagnetic directions at the sites unreliable, and they were excluded from the statistical analysis which gave the formation mean directions.

Interpretation of the formation mean direction by direct comparison with paleomagnetic data from Europe and Africa is hampered by the scarcity of acceptable data from these continents. Although eight previous studies of early Tertiary rocks in Europe pass stringent tests of acceptability (proper magnetic cleaning, well-dated formations, site locations in tectonically undisturbed regions), only two suitable studies have been made in early Tertiary African rocks and two studies in Late Cretaceous European rocks; no satisfactory studies have been made in Late Cretaceous African rocks, and there are only two studies in Early Cretaceous rocks [*Van der Voo and French,* 1974].

A superior technique, which utilizes all the available paleomagnetic data of a given geologic period for the Atlantic-bordering continents, consists first of determining the relative position of each continent according to a model of the opening of the Atlantic Ocean [*Pitman and Talwani,* 1972] and then of combining paleomagnetic data from the different continents to determine a virtual geomagnetic pole (VGP) position. In this way the optimum absolute positions on the globe of the continents at different geologic times may be calculated [*Phillips and Forsyth,* 1972; *Van der Voo and French,* 1974]. From such reconstructions a better estimate of the VGP position may be computed for a particular continent and geologic period which may be poorly represented by paleomagnetic data, as in the case of Africa in the Late Cretaceous.

Following such a technique, *Van der Voo and French* [1974] have determined VGP positions with respect to present coordinates for the Atlantic-bordering continents from late Carboniferous to Eocene. Their VGP positions for Europe and Africa in the Late Cretaceous and early Tertiary enable us to calculate the expected declinations and inclinations at our scaglia rossa sampling locations (Table 3).

Because of our present inability to date most of our sites paleontologically we cannot segregate them according to age to allow close comparison with the predicted inclinations. The formation mean inclination of our entire assemblage of sites, covering a large geographic area and a large interval of time, is 43.7° ± 6.1°. This inclination is in good agreement with the ranges predicted for the sampling area if Italy occupied its present latitude with respect to either Europe or Africa during the deposition of the scaglia rossa.

The most striking feature of the site declinations (Figure 1) is their close parallelism with the northwest-southeast trend of the mountain chain. The formation mean direction has a declination of 317.2°. By comparing this with the anticipated declination of 357.6° to 4.7° derived from the European Late Cretaceous and early Tertiary VGP's it is evident that the scaglia rossa directions are fairly uniformly rotated by 40°–48° counterclockwise relative to the stable European plate. This rotation is in the same sense and of comparable magnitude to rotations observed elsewhere in the western Mediterranean, but this is the first such evidence from the Italian peninsula.

Preliminary results based upon a large number of samples representing the entire scaglia rossa indicated an average counterclockwise rotation of the Italian peninsula of approximately 40° [*Lowrie and Alvarez,* 1974]. In a subsequent preliminary report, *Premoli Silva et al.* [1975] have suggested a counterclockwise rotation of almost 90° for lower Tertiary scaglia rossa rocks alone. However, their results were based upon a limited number of samples (39) from the most weakly magnetized part of the section at Gubbio; further work has indicated that these particular samples were overprinted by a hard secondary magnetization and that the suggested 90° rotation is not valid (G. Napoleone and A. G. Fischer, personal communication, 1974).

Interpretation of the origin of the bend in the Umbrian Apennines is hindered by the present lack of paleontological dating at most sites. Some of the most northern sites apparently have more westerly declinations than sites further to the south along the curve of the Apennines. The angular deviation is not consistent around the arc. The declinations at sites

TABLE 3. Mean Directions of Magnetization of Groups of Sites and Their Corresponding VGP Positions

Data Group	N	Mean Direction of Magnetization		k	α_{95}	VGP Position	
		Declination	Inclination			Latitude	Longitude
Bottaccione gorge							
Early Tertiary and late Maestrichtian sites*	6	338.1	37.3	76.8	7.7	61.3°N	120.5°W
Early Campanian to Turonian sites†	6	286.0	40.0	82.5	7.4	26.7°N	69.8°W
All SR13 sites	16	311.2	41.6	14.2	10.1	45.8°N	86.4°W
Geographically distributed sites, excluding SR13	18	322.7	45.3	22.4	7.5	55.8°N	92.9°W
All scaglia rossa sites	34	317.2	43.7	17.2	6.1	51.1°N	89.5°W
Europe							
Early Tertiary	•••	4.7§	51.6§	•••	•••	78.6°N¶	172.3°E¶
Late Cretaceous	•••	357.6§	43.1§	•••	•••	72.0°N¶	159.6°W¶
Africa							
Early Tertiary	•••	354.7§	48.0§	•••	•••	75.4°N¶	148.3°W¶
Late Cretaceous	•••	342.2§	35.1§	•••	•••	62.0°N¶	129.0°W¶

k is Fisher's precision parameter; α_{95} is radius of circle of confidence.
*Sites SR13A, SR13F, SR13G, SR13B, SR13L, and SR13H.
†Sites SR13N, SR13D, SR13O, SR13J, SR13P, and SR13E.
§Magnetization directions predicted for the sampling area. These values correspond to the VGP positions of *Van der Voo and French* [1974].
¶VGP positions of *Van der Voo and French* [1974] for Europe and Africa.

SR14, SR15, SR17, and SR18 are some 45° further to the west than the remaining sites to the south; however, site SR16 does not agree with this pattern (Figure 1). Some of the angular change is probably attributable to the rotation of peninsular Italy during deposition of the limestone. Without better dating of each individual site we cannot use the results of the stratigraphically distributed sampling to correct for declination variation due to rotation during deposition (see following section).

With the exceptions mentioned above, the site declinations show no systematic deflections attributable to oroclinal bending after deposition. It is possible that the arcuate bending may have been achieved by en échelon transportation and simultaneous rotation of small crustal blocks about subvertical axes as suggested for the northern Appalachians [*Knowles and Opdyke,* 1968]. However, there is no evidence for this type of tectonic movement, and the mechanism has never been suggested for the Umbrian Apennines.

Results of Stratigraphically Distributed Sampling

Stratigraphically distributed sampling was needed to test whether any of the rotation of Italy relative to Europe took place during the approximately 50 m.y. represented by the formation and to establish the paleomagnetic reversal stratigraphy.

Bottaccione section at Gubbio. The town of Gubbio (Figure 1) lies at the foot of a range of low mountains in which the Jurassic to Tertiary Umbrian sequence dips homoclinally to the northeast. Several stream valleys cut through this range and debouch in the valley of Gubbio. One of these gorges carries the main road and a medieval aqueduct, the 'Bottaccione.'

In the Bottaccione gorge the scaglia rossa is well exposed, evenly dipping, and free from major faulting or other severe structural complications. This has been a classic Upper Cretaceous–lower Tertiary section since the pioneering studies of foraminifera in thin section by *Renz* [1936]. The Bottaccione section achieved further importance when *Luterbacher and Premoli Silva* [1962, 1964] showed that the rock record was complete across the Cretaceous-Tertiary boundary. At most other places in the formation and, indeed, in the world, one or more foraminiferal zones are missing in the lowermost Paleocene.

Sampling in the Bottaccione. The Bottaccione section was measured to locate the paleomagnetic sample sites. Thicknesses of the various stratigraphic intervals differed by only a few percent from those given by *Renz* [1936]. The absolute age for each site (Figure 8) was found by assigning the ages of the Phanerozoic time scale [*Quarterly Journal of the Geological Society of London,* 1964] to the paleontological boundaries of *Renz* [1936] and assuming constant deposition rates between these points. The Phanerozoic time scale simply assigned equal time intervals to the six Late Cretaceous stages. This was done for convenience in the absence of better data, but there is still no agreement on an improved time scale.

Two difficulties with the dating should be noted. First, the stratigraphic thickness assigned to the Maestrichtian is disproportionately large, based on any of the proposed absolute time scales for the Cretaceous [*Baldwin et al.,* 1974]. This suggests a high sedimentation rate during the Maestrichtian, although there is no obvious difference in lithology. Second, *Renz* [1936] was not able to separate the four stages of the Turonian to the Campanian.

In view of these problems it is clear that the absolute ages given in Figure 8 are open to considerable question, particularly for the Upper Cretaceous sites. There is, on the other hand, no question of the stratigraphic order of the sites.

Remanent magnetization directions. Nine sites in the Bottaccione gorge were found to be normally magnetized, and six were reversely magnetized. At one site (SR13K), unstable remanent magnetizations were measured; the samples from this site had a large low-coercivity fraction which made them unusually susceptible to isothermal, viscous, and anhysteretic secondary magnetizations. These effects prohibited the attainment of satisfactory magnetic cleaning, and the data from this site were therefore excluded from further analysis.

Both normal and reversed directions were measured at site 13A and could be separated according to their stratigraphic position. Six negative directions at this site were clustered around the antipodal direction to the normal group, and six gave intermediate directions. It could not be established whether or not the latter represented real intermediate geomagnetic field directions, and it was necessary to exclude them from site statistical analysis which was based on the remaining 13 (Table 1).

In Figure 8 the declinations and inclinations of the 16 Bottaccione sites are plotted against stratigraphic position and approximate absolute age. The figure also shows the error bars for these directions. The error in inclination is equal to the radius of the circle of confidence (α_{95}); the declination error is equal to $\alpha_{95}/\cos I$.

The mean direction of the Bottaccione sites was $D = 311.2°$ and $I = 41.6°$ (Table 2). This direction is only 4.9° different from the mean direction of all the scaglia rossa sites.

The results show no significant change in inclination through the formation (Figure 8). An inclination of 38° passes through the error bars of all 16 sites, suggesting that the site of deposition stayed at an approximately constant paleolatitude during this time interval.

The declinations of the Bottaccione sites show well-defined trends. The declination of the mean of early Campanian and older sites is 72° west of the European Late Cretaceous direction, while that of the early Tertiary and late Maestrichtian sites is 27° west of the European early Tertiary direction (Table 3). There are therefore two significant declination changes: the first consists of a shift of about 45° counterclockwise during deposition in the Campanian-Maestrichtian, and the second corresponds to an additional change of about 27° in the same sense subsequent to deposition.

The pattern of declinations observed in the Bottaccione section could be produced either (1) by two episodes of counterclockwise rotation of the Italian lithosphere, one during and the other after deposition of the scaglia rossa, or (2) by late local tectonic effects, in particular by physical rotation of adjacent beds, distributed over certain stratigraphic intervals, resulting from the deformations that produced the folds of the Umbrian Apennines. It would be virtually impossible to see the effects of the second mechanism by the techniques of structural geology. One or more paleomagnetic control sections through the scaglia rossa, outside the range of local tectonic effects, will be required to establish whether the mechanism was or was not active. In view of the absence of any other reason for suspecting that local interbed rotation was active, we feel justified in considering (in the discussion) some of the consequences of the possible history of tectonic rotations of the Italian lithosphere suggested by the Bottaccione data.

It should be noted that here we are trying to establish the fine details of the rotational history. Even if it should later be

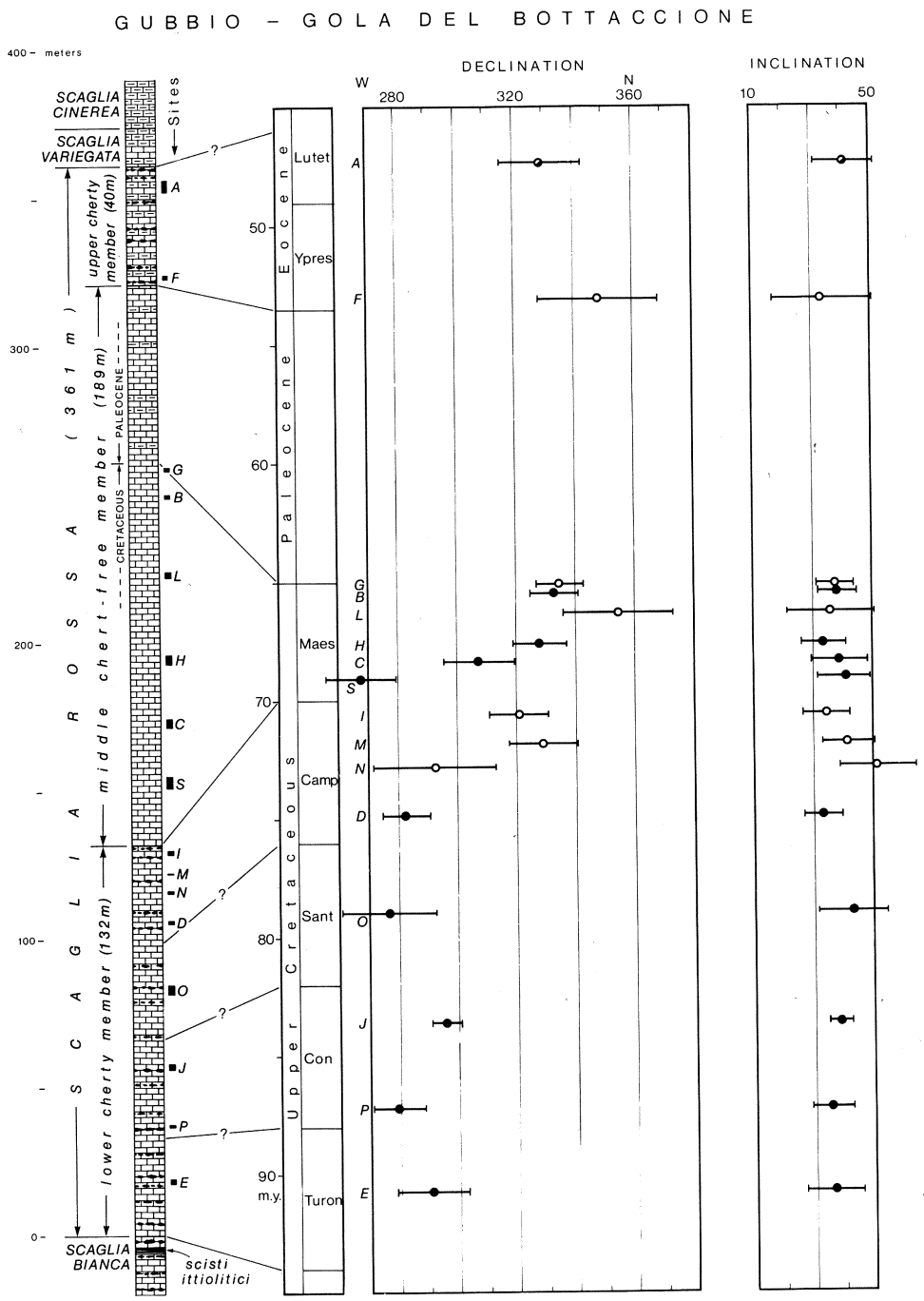


Fig. 8. Paleomagnetic stratigraphy of the scaglia rossa in the Bottaccione gorge section at Gubbio. First column is schematic lithostratigraphy with contacts picked by E. Centamore (personal communication, 1973) and thicknesses measured during this study. Paleocene-Cretaceous boundary is from *Luterbacher and Premoli Silva* [1962, 1964]. The stratigraphic intervals represented by paleomagnetic sites SR13A through SR13S are indicated. Second column is absolute age assignments. Tertiary time scale is from *Berggren* [1972]; Cretaceous time scale is from the *Quarterly Journal of the Geological Society of London* [1964]. Solid tie lines to the first column are based on *Renz*'s [1936] paleontologic stage boundaries for the Gubbio sections. Third column is declinations of paleomagnetic site mean directions (solid circles, normal; open circles, reversed; site SR13A, mixed polarities). Declination error is obtained from within-site statistical analysis; $\delta D = \alpha_{95}/\cos I$. Fourth column is site mean inclinations; the inclination error is equal to α_{95}.

shown that local effects are responsible for part of the Bottaccione pattern, this would not invalidate the evidence for a rotation of the Italian lithosphere derived from the geographically distributed sampling.

Paleomagnetic stratigraphy. The distribution of normal and reversed sites in the Bottaccione section is shown in Figure 8. The uncertainties in dating are a problem, and many gaps remain to be filled in, but we have been able to establish some of the gross features of the reversal stratigraphy from our 16 sites (Figure 9). These agree well with the Late Cretaceous reversal pattern proposed by *Larson* [1974, also personal communication, 1974].

Site 13G, immediately below the Cretaceous-Tertiary boundary, is reversed, in agreement with the reversed interval at that boundary shown by Larson. Four sites in the rest of the Maestrichtian (13B, 13H, 13C, and 13S) are normal;

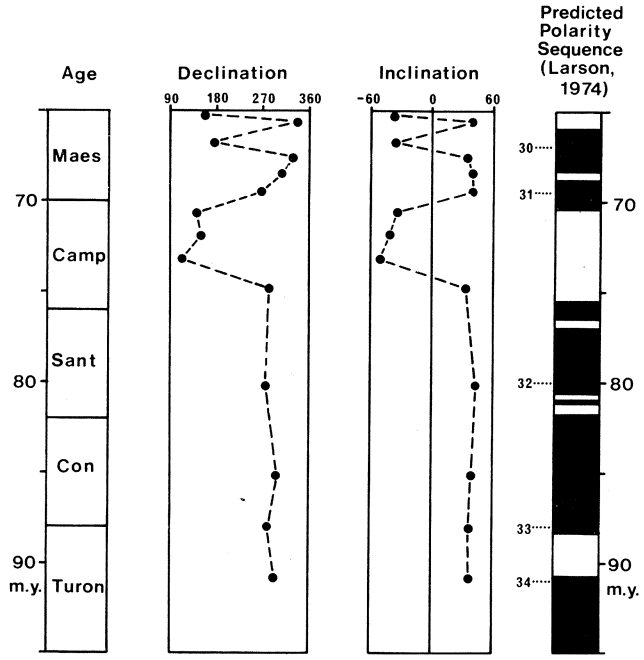


Fig. 9. Comparison of the magnetic stratigraphy of the scaglia rossa formation in the Bottaccione gorge with the Late Cretaceous geomagnetic polarity sequence deduced from oceanic magnetic anomalies by *Larson* [1974].

these correspond to the largely normal interval of anomalies 30–31, which occupy most of the Maestrichtian in Larson's scale. Reversed site 13L would then correspond to the short reversed interval between anomalies 30 and 31.

Sites 13I, 13M, and 13N mark the upper Campanian as reversed, again in agreement with Larson. The polarities of our older sites (13D, 13O, 13J, 13P, and 13E) are entirely normal and agree with Larson's largely normal sequence below the middle Campanian. Close sampling would be needed to locate the smaller reversed zones in this interval; we would expect to find the reversed interval of anomalies 33–34 near the base of the formation. A more rapid reversal sequence and the low magnetization intensities in the Tertiary scaglia rossa make it a more difficult problem to establish a reversal stratigraphy in the upper part of the formation.

DISCUSSION

The paleomagnetic data from all scaglia rossa sites give a VGP at 51°N, 90°W. In view of the long period of deposition represented by the formation (Turonian to middle Eocene, approximately 50 m.y.) this VGP position averages paleomagnetic data which span a long time interval in which appreciable continental drift or polar wander may have taken place and should be considered with caution. In the absence of confident paleontological dates for most of the geographically distributed sites we cannot at present separate the early Tertiary paleomagnetic directions from those that are older.

There is some indication of a progressive change in declination from a northerly direction in the south of our sampling area to a west-northwest direction at some of the sites in the north (Figure 1). Again, because of lack of paleontologic dating we cannot positively attribute this to a tectonic modification, as might ensue from oroclinal bending. If, as indicated by the data from the Bottaccione gorge, the Italian peninsula was rotating during the depositional period of the scaglia rossa, the apparent declination variation may be an artifact of the distribution of ages of the sampling sites.

The data from the Bottaccione gorge provide the only means at present of separating the paleomagnetic directions according to geologic age. Although the precise dating of the Bottaccione sites may be inexact, their stratigraphic order is not in question. We may therefore separate these 16 sites into three groups: (1) an upper group comprising the lower Tertiary and upper Maestrichtian sites (13A, 13F, 13G, 13B, 13L, and 13H), (2) a middle group comprising the lower Maestrichtian and upper Campanian sites (13C, 13S, 13I, and 13M), and (3) a lower group comprising the remaining Upper Cretaceous sites (13N, 13D, 13O, 13J, 13P, and 13E). These groups correspond to paleomagnetic directions acquired after, during, and before the proposed Campanian-Maestrichtian episode of rotation. Of primary interest are the paleomagnetic directions of the upper and lower groups, acquired while the deposition was taking place during essentially stationary conditions.

The mean direction of the six sites in the upper group corresponds to a VGP at 61°N, 121°W, while that of the three (stationary) Upper Cretaceous sites corresponds to a VGP at 27°N, 70°W (Table 3). These VGP positions may be immediately compared to early Tertiary and Cretaceous VGP positions for Europe and Africa (Table 3).

The early Tertiary VGP positions from European and African data are close to each other but quite far from the Bottaccione early Tertiary VGP (Figure 10). The mean European Late Cretaceous VGP and the African Late Cretaceous VGP [*Van der Voo and French,* 1974] are quite separate, as a result of relative plate motions since the Early Cretaceous. Paleomagnetic data from Sicily, which gave a Cretaceous VGP near that of Africa, have been used to support the concept that Sicily is a part of the African plate [*Barberi et al.,* 1974]. On the other hand, the scaglia rossa Late Cretaceous pole is quite far from both the African and the stable European VGP positions. From this we conclude that the northern Apennine region of the Italian peninsula did not belong to either the African or the stable European plate from the Late Cretaceous to the middle Eocene but moved in a separate, although not necessarily independent, manner.

Our model for the motion of the Italian peninsula, based primarily on the scaglia rossa data from the Bottaccione gorge

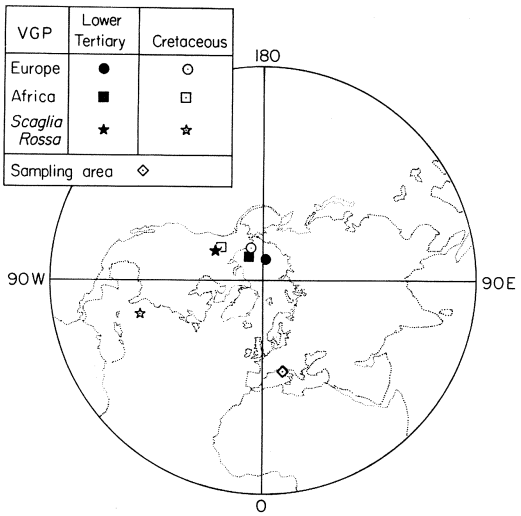


Fig. 10. The VGP positions for the lower Tertiary and Cretaceous from Europe and Africa [*Van der Voo and French,* 1974] and the scaglia rossa (this study).

and to be considered tentative until the measurement of one or more check sections, shows no rotation from 95 to 75 m.y. B.P., a 45° counterclockwise rotation between 75 and 65 m.y. B.P., no rotation from 65 to 45 m.y. B.P., and a 25° counterclockwise rotation at some time after 45 m.y. B.P.

The separation of North America from Europe and Africa since the Late Triassic has been inferred from magnetic anomaly studies in the North Atlantic [*Pitman and Talwani,* 1972]. *Smith* [1971] and *Dewey et al.* [1973] showed that plate movements in the Mediterranean are related to the spreading phases in the Atlantic.

Three phases of Eurasia-Africa motion were recognized by *Dewey et al.* [1973], using the rotations and poles of *Pitman and Talwani* [1972]. The first phase began at about 180 m.y. B.P. with separation of North America and Africa, producing left lateral strike slip motion between Eurasia and Africa. The second phase began at about 80 m.y. B.P. when Eurasia separated from North America and moved east faster than Africa, so that right lateral strike slip motion occurred between Eurasia and Africa. In the third phase, which began at about 53 m.y. B.P., Eurasia and Africa were moving east away from North America at about the same rate and showed a slight north-south convergence. A generally similar, three-phase movement history was inferred by *Smith* [1971].

There appears to be an approximate coincidence in timing between the second phase in the Atlantic opening model (80–53 m.y.) and the 45° counterclockwise rotation of Late Cretaceous age in our Bottaccione model (75–65 m.y.). The coincidence is still closer between the Bottaccione rotation and the first of two subphases (80–63 m.y.) that *Dewey et al.* [1973] distinguish in their second phase. There are large margins of error in dating both the Bottaccione rotation and the changes in the Atlantic spreading pattern, and improvements in dating could either strengthen or weaken the coincidence in timing. We suggest, however, that both the beginning and the end of the Late Cretaceous rotation of peninsular Italy shown by the declination change in the scaglia rossa of the Bottaccione section may be the result of changes in the spreading pattern of the Atlantic Ocean.

During the proposed Campanian-Maestrichtian (75–65 m.y. B.P.) episode of rotation of the Italian peninsula the general motion of Africa was northwest in relation to Europe, and since the Eocene it has been primarily northward. It is not possible to ascribe the counterclockwise rotation of Italy to a 'ball bearing effect' resulting from the shear component of the motion of the two plates, which, at the time, was right lateral and would have produced rotation in the opposite sense to that observed. The rotation must be a result of microplate adjustment to the predominantly compressive, Late Cretaceous–Tertiary motion between Africa and Europe.

CONCLUSIONS

The large body of paleomagnetic data from the scaglia rossa resolved satisfactorily some of the principal aims of the study, while it left other problems only partly answered.

1. The limestone has a weak stable magnetization, carried either by magnetite or hematite in very fine grain sizes; both are present at many sites. Secondary magnetization components were apparently acquired sufficiently soon after the primary magnetization for them to be considered essentially contemporaneous with the time of deposition, although this conclusion may not necessarily be valid if extremely close stratigraphic sampling is done. The directions of the samples measured in this study give meaningful paleomagnetic information about conditions during the lengthy deposition of the formation.

2. The limestone magnetization direction probably does not have any depositional inclination error. The observed inclinations are in good agreement with expected values for the sampling site according to the plate reconstruction model of *Van der Voo and French* [1974].

3. The declinations at sites distributed over the outcrop area show no definitive systematic variation that can be used incontrovertibly in support of the hypothesis of oroclinal bending in the northern Apennines, although some local bending cannot be ruled out.

4. The mean formation direction requires an approximately 45° counterclockwise rotation of the Italian peninsula since the mid-Cretaceous. Our tentative conclusion from sites placed at different stratigraphic levels where the entire section is exposed is that the total rotation may be as much as 70° and that it took place in two distinct episodes. The first of these is thought to be a Campanian-Maestrichtian rotation of about 45°, followed by a later post–middle Eocene rotation of about 25°. Confirmation of the present picture will require paleomagnetic studies in other complete sections through the formation.

5. The Late Cretaceous rotation (approximately 75–65 m.y. B.P.) appears to coincide with one of the three principal phases of sea floor spreading in the Atlantic. The amount and sense of rotation indicates that the Italian peninsula was not attached either to Eurasia or to Africa, nor did it behave as a ball bearing caught between the two major plates. It seems to have moved as an independent microplate.

6. The magnetic polarities of the sites agree well with the most recent pattern of polarity intervals of *Larson* [1974]. Detailed investigation, by means of more closely placed sites in the Gubbio gorge and in parallel sections, will refine the magnetic stratigraphy and provide a check on the sea floor spreading polarity sequence.

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